Regularization, Stability, and Generalization

Rhodine Bendict Etyang

May 7, 2026

Abstract

This project dives into the relationship between regularization, algorithmic stability, and generalization in supervised learning . Ridge regression is implemented from scratch and evaluated across different values of the regularization parameter α, with ordinary least squares corresponding to the case $\alpha = 0$. Using a leave-one-out procedure, Stability is measured . The aim is to study the resulting change in predictions on a fixed test set . The results show that increasing regularization improves the stability of the algorithm by reducing the sensitivity of the model to the training data. Test error initially decreases with regularization upto a certain point ,beyond which it increases for very large values of α due to underfitting. There is a negative correlation between stability and test error , providing evidence that more stable models tend to generalize better. Finally,the experiment also studies the effect of training dataset size, it proves that larger datasets lead to increased stability and improved predictive performance.

1 Introduction

A fundamental challenge in supervised learning lies in the gap between *Empirical Risk Minimization* (ERM), which focuses on minimizing the loss on the observed training data, and *generalization*, which concerns the ability of a model to perform well on unseen data. Classical learning theory addresses this gap using tools such as uniform convergence and complexity measures, which relate the richness of a hypothesis class to its generalization performance. In this framework, simpler models are expected to generalize better.

However, this perspective becomes less informative in high-dimensional or overparameterized settings. In such cases, complexity measures can be very large, leading classical theory to predict poor generalization, even when models perform well empirically. This suggests that model complexity alone does not fully explain generalization.

In this work, the focus is shifted from the structure of the hypothesis class to the behavior of the learning algorithm itself. Following the framework proposed by Bousquet and Elisseeff, generalization is studied through the lens of *algorithmic stability*. An algorithm is considered stable if small changes in the training data do not significantly affect its predictions.

The objective of this project is to empirically investigate the relationship between regularization, stability, and generalization in the context of ridge regression. By introducing an ℓ_2 regularization term, ridge regression shrinks the magnitude of the model coefficients, reducing sensitivity to individual observations and acting as a stabilizing mechanism.

The remainder of this paper is organized as follows. Section 2 describes the dataset and the preprocessing procedures. Section 3 introduces the models, the stability measure, and the experimental setup. Section 4 presents the experimental results, including the effects of regularization and dataset size. Section 5 discusses the implications of the findings, and Section 6 concludes the study.

2 Dataset

The experiments are conducted using the *Communities and Crime* dataset from the UCI Machine Learning Repository. This dataset is particularly interesting because it contains a large number of features capturing socio-economic characteristics, many of which are highly correlated. For example, variables related to income (such as median household income and poverty rates), employment (unemployment and employment rates), housing conditions, and education levels tend to measure closely related aspects of community structure. In such settings, the matrix $X^T X$ is often nearly singular, which makes ordinary least squares estimates unstable and highly sensitive to small perturbations in the data. This makes the dataset particularly suitable for studying the stabilizing effect of regularization.

Several preprocessing steps were applied to ensure data quality. First, non-predictive identifier columns such as state, county, and community name were removed. Next, columns with more than 50% missing values were discarded, as they contained insufficient information. The remaining missing values were handled using mean imputation, where missing entries were replaced with the average value of the corresponding feature. Finally, all features were standardized using the mean and standard deviation computed on the training set, ensuring that all variables are on a comparable scale and preventing features with large magnitudes from dominating the model.

After preprocessing, the dataset consists of 1994 observations and approximately 100 features. The target variable is the *violent crime rate per population*, which represents the level of violent crime within each community. It is a continuous variable normalized between 0 and 1, making the problem suitable for regression analysis.

3 Methods

3.1 Models

Two linear models are considered in this study: ordinary least squares (OLS) and ridge regression. Both models were implemented from scratch using NumPy.

3.1.1 Ordinary Least Squares (OLS)

The OLS model serves as the unregularized baseline. Given a design matrix $S \in \mathbb{R}^{n \times d}$ and a target vector $y \in \mathbb{R}^n$, the parameters are estimated by minimizing the sum of squared residuals.

The closed-form solution is given by:

$$w = (S^T S)^{-1} S^T y$$

To ensure numerical stability, particularly when the matrix $S^T S$ is singular or nearly singular, the standard matrix inverse is replaced with the Moore–Penrose pseudo-inverse. In practice, this is implemented using `np.linalg.pinv`.

The implementation follows three main steps: first, the Gram matrix $S^T S$ is computed; second, the pseudo-inverse is applied; and finally, the result is multiplied by $S^T y$ to obtain the parameter vector w.

Predictions are then obtained as:

$$\hat{y} = Sw$$

3.1.2 Ridge Regression

Ridge regression extends the OLS formulation by introducing an ℓ_2 regularization term to the objective function.

The regularized solution is given by:

$$w = (S^T S + \alpha I)^{-1} S^T y$$

where $\alpha \geq 0$ is the regularization parameter.

In the implementation, the identity matrix I is modified to ensure that the intercept term is not penalized. This is achieved by setting the corresponding diagonal entry to zero.

The training procedure follows similar steps to OLS. First, the matrix $S^T S$ is computed, then the regularization term αI is added, and the resulting matrix is inverted using the pseudo-inverse. The parameter vector is then obtained by multiplying with $S^T y$.

The addition of the regularization term improves the conditioning of the matrix $S^T S$, resulting in a more stable solution. In particular, it reduces the sensitivity of the model to small perturbations in the training data, which is especially important in high-dimensional settings.

3.2 Stability Measure

Algorithmic stability measures how sensitive a learning algorithm is to small changes in the training data. Intuitively, a stable algorithm produces similar predictions even when the training set is slightly modified, whereas an unstable algorithm may produce significantly different outputs when a single observation is removed.

In this study, stability is estimated using a leave-one-out procedure. For each training observation, the model is retrained after removing that observation, and the resulting predictions on a fixed test set are compared to the original predictions obtained using the full training set.

The stability metric is defined as the average absolute change in predictions:

$$\text{Stability} = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{m} \sum_{j=1}^{m} \left| \hat{y}_j^{(-i)} - \hat{y}_j \right|$$

where $\hat{y}_j^{(-i)}$ denotes the prediction obtained after removing the i-th training point, n is the number of training observations, and m is the number of test observations.

Lower values of this metric indicate that the model is less sensitive to individual training points and is therefore more stable. This notion of stability is closely related to generalization: algorithms that are less sensitive to small perturbations in the training data tend to produce more consistent predictions on unseen data, leading to improved generalization performance.

3.3 Experimental Setup

The dataset is split into training and test sets using an 80/20 random split, ensuring that the evaluation of model performance is conducted on unseen data. All features are standardized using the mean and standard deviation computed on the training set, and the same transformation is then applied to the test set. An intercept term is added to the design matrix by augmenting it with a column of ones.

Models are trained for different values of the regularization parameter:

$$\alpha \in \{0, 0.001, 0.01, 0.1, 1, 10, 100, 1000\}$$

where $\alpha = 0$ corresponds to ordinary least squares, and larger values correspond to increasing levels of regularization.

In addition to prediction error, stability is computed for each value of α using the procedure described in the previous section. For each model, the stability score is calculated by measuring the average change in predictions on a fixed test set when individual training observations are removed. This produces a stability value associated with each level of regularization.

To study the relationship between stability and generalization, the stability scores are compared directly with the corresponding test errors across different values of α. In particular, the correlation between stability and test error is computed, and a plot of stability versus test error is generated. This allows us to empirically assess whether more stable models tend to achieve better generalization performance.

Finally, an additional experiment is conducted to examine the effect of training dataset size. For a fixed value of α, models are trained on progressively larger subsets of the training data. For each subset, both test error and stability are computed. This provides further insight into how the amount of available data influences model stability and predictive performance.

4 Experiments and Results

4.1 Error vs Regularization

Figure 1 shows the training and test error as a function of the regularization parameter. For small values of α, ridge regression behaves similarly to OLS. As α increases, test error initially decreases, indicating improved generalization, reaching its minimum at $\alpha = 10$. For very large values of α, both training and test error increase, reflecting underfitting.

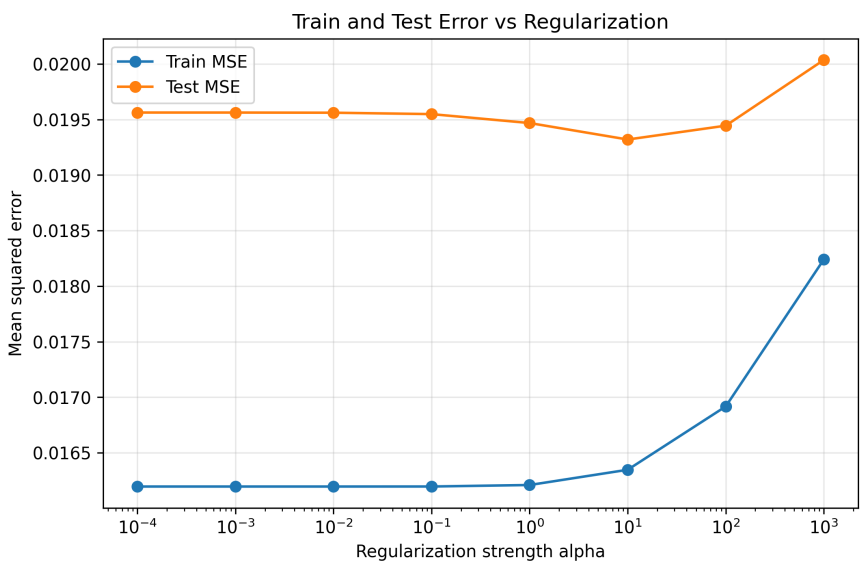

Figure 1: Training and test error vs regularization strength

4.2 Stability vs Regularization

Figure 2 shows the inverse relationship between regularisation strength and prediction sensitivity. For higher values of α,the model becomes significantly less sensitive to individual training points

indicating higher algorithmic stability .Since stability is measured as prediction change, lower values indicate more stable models.

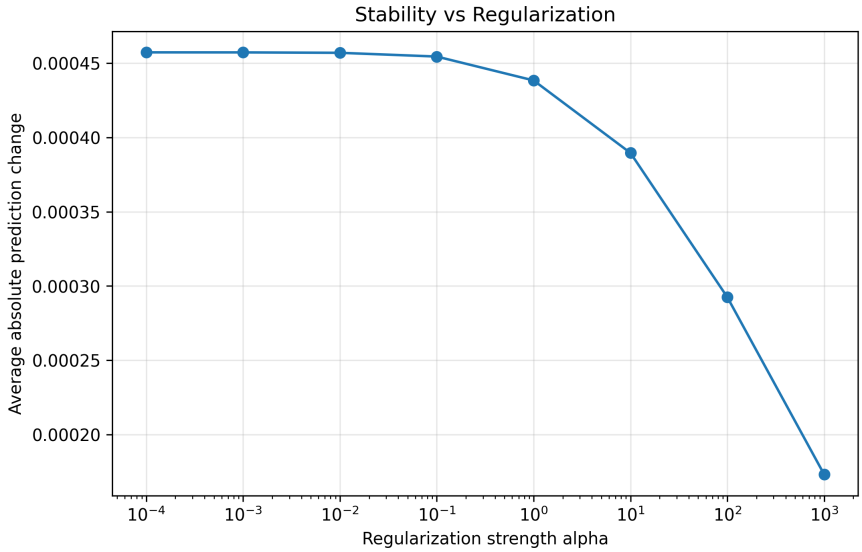

Figure 2: Stability vs regularization strength

4.3 Stability vs Test Error

Figure 3 illustrates the relationship between stability and generalization. It reveals a characteristic U-shaped curve that maps how the model transitions through the bias–variance tradeoff as the regularization strength (α) changes:

- **Overfitting (Left Side):** At low α values (e.g., 0.001 to 0.1), the model is highly sensitive to the training data (unstable) and exhibits higher test error.
- **Optimal Generalization (Bottom):** The point at $\alpha = 10$ represents the "sweet spot," where the model achieves the lowest test MSE by balancing stability and flexibility.
- **Underfitting (Right Side):** At extreme regularization ($\alpha = 1000$), the model becomes highly stable but loses its predictive power, resulting in a sharp increase in test error.

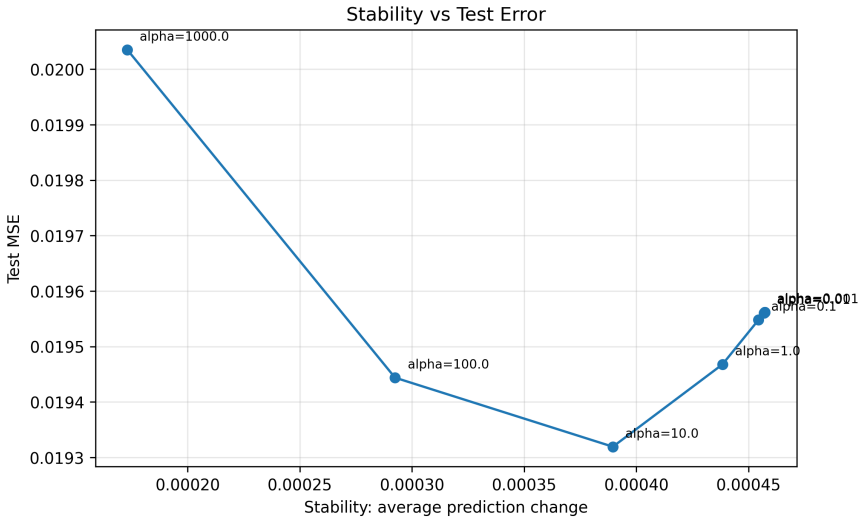

Figure 3: Relationship between stability and test error across different values of α

4.4 Effect of Dataset Size

In addition to studying the effect of regularization, it is important to understand how the size of the training dataset influences stability. Intuitively, as the number of training observations increases, the influence of any single data point on the learned model decreases. This suggests that larger datasets may naturally lead to more stable learning algorithms.

In small datasets, individual observations, especially noise or outliers ,can have a significant impact on the estimated parameters, resulting in unstable predictions. In contrast, as the dataset grows, the effect of individual perturbations is averaged out, leading to more consistent and robust predictions. This behavior is closely related to the notion of consistency in learning algorithms, where performance improves as more data becomes available.

The Stability vs. Training Set Size plot demonstrates this visually. stability is a function of data volume. As the training set grows, the algorithm becomes increasingly robust, with the average prediction change decreasing significantly before eventually stabilizing. This confirms that larger datasets provide a natural stabilizing effect . α.

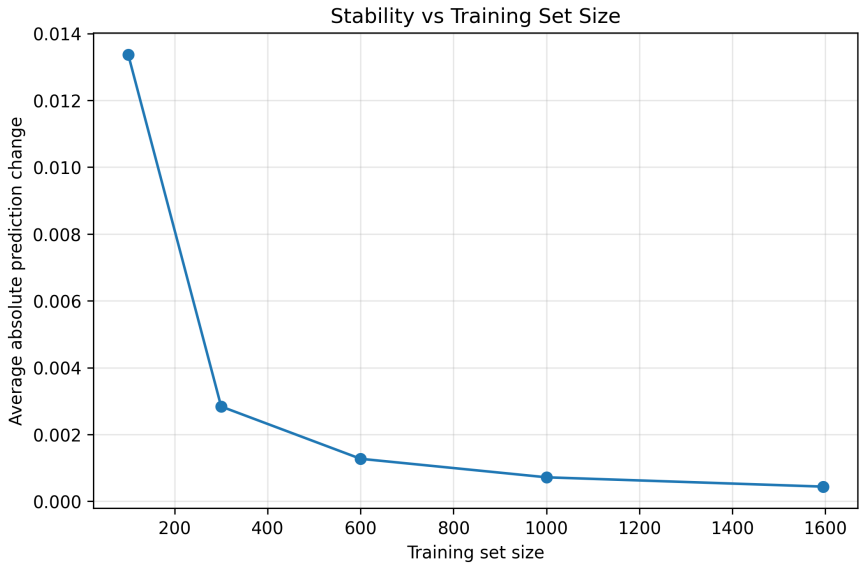

Figure 4: Stability as a function of training set size

5 Discussion

One of the central expectations of this study is that increasing the regularization parameter α should improve the stability of the learning algorithm. This is clearly supported by the results. As α increases, the stability metric measured as the average change in predictions decreases significantly, indicating that the model becomes less sensitive to individual training observations.

This behavior can be explained by examining the effect of regularization on the underlying optimization problem. By adding the term αI to the matrix $S^T S$, ridge regression effectively constrains the magnitude of the coefficients. This reduces the model's flexibility and prevents it from reacting strongly to small variations in the training data. As a result, the learned parameters depend more on the overall structure of the data rather than on individual observations or noise.

Empirically, this is reflected in the downward trend observed in the stability plot. For small values of α, the model behaves similarly to ordinary least squares and exhibits relatively high instability. As α increases, the average prediction change drops sharply, demonstrating a clear stabilizing effect. By the time α reaches large values (e.g., $\alpha = 1000$), the model becomes highly stable, with very small changes in predictions when individual training points are removed.

However, this increased stability comes at a cost. While the model becomes more robust, excessive regularization reduces its ability to capture meaningful patterns in the data, leading to underfitting. Another key objective of this experiment is to understand how stability relates to generalization performance. The results provide clear evidence of a strong connection between the two. The optimal balance is achieved at $\alpha = 10$, where the model attains its lowest test error. At this point, the model is sufficiently stable to ignore noise in the data while remaining flexible enough to capture the underlying structure. This corresponds to an optimal tradeoff between bias and variance, where neither underfitting nor overfitting dominates.

Overall, demonstrates that generalization is not a monotonic function of stability. While increased stability can improve generalization up to a point, excessive stability leads to a loss of predictive power. This highlights that stability acts as a proxy for the bias variance tradeoff: low stability corresponds to high variance, while high stability corresponds to high bias. The ob-

served U-shaped relationship confirms the existence of an optimal level of stability that minimizes generalization error.

6 Conclusion

This study validates the fundamental link between regularization, algorithmic stability, and generalization through the implementation of ordinary least squares (OLS) and ridge regression. By analyzing the model's behavior across varying levels of α and different training dataset sizes, several key conclusions emerge:

- **Regularization as a Stabilizer:** The introduction of the ℓ_2 penalty term (αI) directly improves algorithmic stability by constraining the magnitude of the coefficients, as evidenced by the sharp decline in prediction sensitivity as α increases.
- **Optimal Stability for Generalization:** The results confirm that generalization is not achieved through infinite stability, but through an optimal balance. The lowest test MSE is observed at a moderate level of stability (at $\alpha = 10$), reflecting an effective tradeoff between bias and variance.
- **Effect of Dataset Size :** Beyond regularization, increasing the size of the training dataset provides a natural stabilizing effect. As more data becomes available, the model relies less on individual observations, leading to improved robustness and a reduced generalization gap.

Declaration

I declare that this material, which I now submit for assessment, is entirely my own work and has not been taken from the work of others, save and to the extent that such work has been cited and acknowledged within the text of my work. I understand that plagiarism, collusion, and copying are grave and serious offences in the university and accept the penalties that would be imposed should I engage in plagiarism, collusion or copying. This assignment, or any part of it, has not been previously submitted by me or any other person for assessment on this or any other course of study.

7 References

1. O. Bousquet and A. Elisseeff, "Stability and Generalization," *Journal of Machine Learning Research*, vol. 2, pp. 499–526, 2002.
2. Communities and Crime Dataset, UCI Machine Learning Repository. Available at: `https://archive.ics.uci.edu/ml/datasets/communities+and+crime`
3. T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, Springer, 2009.